

Equivalence Partitioning

A design technique that divides inputs or conditions into classes expected to behave equivalently.

Idea

Instead of testing every possible value, group valid and invalid values that should be handled in the same way.

Example

For an accepted age from 18 to 65: classes can be <18, 18–65 and >65.

Benefit

Reduces combinations while maintaining representativeness, especially when the input domain is large.

Cautions

Classes need to reflect real rules; poorly defined partitions can hide important differences.

When to use

Use this knowledge when risk, architecture or the development lifecycle stage requires specific validation. Before applying it, confirm the objective, environment, required data and expected evidence.

How QA applies it in practice

Start from the behavior or risk that needs validation, prepare data and environment, execute the technique or tool reproducibly and record evidence that allows the team to understand the result.